

Evaluating an Offline Detector Against Adversarial Prompt Attacks

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

github.com/poojakira/llm-redteam-framework - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

175

Tests passed

1

Tests skipped

94.30%

Statement coverage

1.00

Random-split F1

SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector's generalization gap explicit instead of hiding it behind a random split.

Verified Evidence

EV-01	Grouped-template F1: 0.9714	EV-02	Novel-phrasing OOD F1: 0.7188
EV-03	OOD precision: 0.5897	EV-04	OOD recall: 0.92
EV-05	OOD false-positive rate: 64% (16/25)	EV-06	175 passed, 1 skipped.
EV-07	94.30% statement coverage; CI gate is 90%.	EV-08	Current benchmark values remain pinned:
EV-09	random split F1 1.00	EV-10	grouped-template F1 0.9714

Method and Control Path

01	Generate adversarial and benign prompt corpora across defined attack categories.
02	Train/evaluate the offline TF-IDF + Logistic Regression detector.
03	Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.
04	Emit JSON/SARIF evidence for CI.
05	Keep live-model behavior claims out of scope unless separately measured.

Limitations and Residual Risk

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.

Reproducibility

Expected CI evidence: 175 passed, 1 skipped, 94.30% coverage.

Evidence Provenance

Claim discipline Metrics come from the repository's verified metrics and research brief.
Scope discipline Benchmarks are reported only at their documented fixture, dataset, or test scope.
Reproduction PDF and PNG are generated locally from committed evidence without paid services.